Comparative Analysis of Kolmogorov-Arnold Networks and Standard Machine Learning Models in Empirical Asset Pricing

Master in Quantitative Economics

University of California, Los Angeles

Kanupriya Parashar

Denis Chetverikov

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Abstract

Motivated by the empirical asset pricing framework of (Gu, Kelly, & Xiu, 2020), this study explores the predictive and economic potential of Kolmogorov-Arnold Networks (KANs) compared to other state-of-the-art machine learning models in forecasting cross-sectional stock returns among the "Magnificent 7" large-cap U.S. equities. We construct a monthly dataset using both market and macroeconomic information from Yahoo Finance and FRED, and implement a comparative analysis of KANs, deep neural networks (DNN), recurrent neural networks (RNN), gradient-boosted regression trees (GBRT), and linear regression. Our evaluation combines statistical accuracy—measured by out-of-sample R^2—with direct tests of economic value, including long-short decile portfolio Sharpe ratios and maximum drawdown. By examining model performance in a highly efficient and concentrated market segment, this research highlights the importance of economic backtesting and portfolio-level

analysis when assessing the practical impact of machine learning models in asset pricing. The findings offer new insights into the capabilities and limitations of advanced nonlinear methods for investment decision-making in modern equity markets.

Keywords: Kolmogorov-Arnold Networks (KAN); Machine Learning; Empirical Asset Pricing; Cross-Sectional Returns; Deep Neural Networks (DNN); Recurrent Neural Networks (RNN); Gradient-Boosted Regression Trees (GBRT); Sharpe Ratio; Drawdown; Long-Short Portfolio; Magnificent 7 Stocks; Macro-Finance; Predictability; Portfolio Backtesting

Introduction

Machine learning has become a transformative tool in empirical asset pricing, challenging long-standing assumptions about the predictability of cross-sectional stock returns. Recent studies, most notably (Gu, Kelly, & Xiu, 2020), have shown that advanced machine learning models can systematically extract information from high-dimensional datasets, identifying return patterns that traditional linear models may overlook. Their seminal work applied a wide array of machine learning techniques—including elastic net, random forests, and deep neural networks—to a massive cross-section of U.S. stocks, demonstrating that sophisticated algorithms not only improve predictive accuracy but can also generate economically meaningful investment strategies (Gu, Kelly, & Xiu, 2020).

Building on these insights, this paper explores whether newer machine learning architectures, specifically Kolmogorov-Arnold Networks (KANs), can further enhance return forecasting and portfolio construction. While much of the literature focuses on broad equity universes, my study examines the performance of these methods in a highly concentrated set of U.S.

“Magnificent 7” mega-cap stocks. This focus raises important questions about the efficacy and robustness of machine learning in environments characterized by extreme efficiency, limited firm heterogeneity, and intense investor scrutiny.

Research Motivation

While prior studies, including (Gu, Kelly, & Xiu, 2020), have generally focused on large and diverse equity universes, the question remains: **How well do these techniques perform in an environment dominated by a small number of large, information-rich stocks?** This question is increasingly relevant as the so-called “Magnificent 7”—Apple, Microsoft, Alphabet, Amazon, Meta Platforms, Nvidia, and Tesla—have come to account for a disproportionate share of both U.S. market capitalization and investor attention. Given their prominence and the depth of analysis they attract from the investment community, these stocks present a unique and challenging environment for testing asset pricing models.

First: **Do modern machine learning models, and in particular Kolmogorov-Arnold Networks (KANs), add meaningful value for return prediction in a highly concentrated, information-efficient universe?** This is an important test,

as most empirical studies to date focus on broad, heterogeneous universes where the opportunity to "find alpha" may be larger simply due to greater variation across firms. In contrast, the Magnificent 7 are intensively analyzed, highly liquid, and represent firms at the technological and financial frontier. It is plausible that statistical models, no matter how complex, may have limited value here.

Second: **How do predictive gains, if any, translate into practical portfolio outcomes?** A central theme of (Gu, Kelly, & Xiu, 2020) is the distinction between statistical and economic performance. A model that marginally improves R^2 may still fail to produce portfolios with superior risk-adjusted returns. To address this, we follow the literature in constructing monthly long-short portfolios based on predicted returns, evaluating their performance through Sharpe ratios and maximum drawdown metrics. This economic backtesting approach is essential to understanding whether model outputs have practical value for investors.

Third: **Are patterns observed in the broader stock universe preserved, diminished, or reversed in this mega-cap setting?** The original Kelly paper found that nonlinear models—such as gradient-boosted trees and deep neural networks—substantially outperform linear regressions, both in prediction and

portfolio tests, when applied to thousands of stocks. However, the empirical environment here is different: with only seven stocks, there is little cross-sectional dispersion and a high degree of market efficiency. This setting serves as a stress test for machine learning models, revealing their limitations and potential in "hard mode" asset pricing.

Why These Questions Matter

The answers to these questions matter for several reasons. From a practical perspective, the Magnificent 7 constitute an outsized share of both U.S. and global equity portfolios. Asset managers, quantitative hedge funds, and retail investors alike are heavily exposed to these stocks, making any insight into their return dynamics valuable. From a methodological standpoint, testing advanced machine learning architectures like KANs in such a demanding setting provides evidence about the boundary conditions of these methods. If models work here, they likely work elsewhere; if not, it speaks to the limits of "algorithmic alpha" in the most efficient corners of the market.

Furthermore, the literature is increasingly clear that statistical metrics alone are insufficient. As markets adapt and sources of return premia evolve, only models that survive out-of-sample and generate positive economic value are of

genuine use. (Gu, Kelly, & Xiu, 2020) emphasize this point, advocating for rigorous backtesting and the use of risk-adjusted performance measures in any assessment of model quality.

Data

Stock Price Data: Yahoo Finance

The primary dataset for this research consists of monthly closing prices for the so-called “Magnificent 7” U.S. mega-cap stocks—Apple (AAPL), Microsoft (MSFT), Alphabet (GOOGL), Amazon (AMZN), Meta Platforms (META), Nvidia (NVDA), and Tesla (TSLA). Data were sourced from Yahoo Finance using the yfinance Python package. This platform was selected due to its broad coverage, reliability, and ease of automated data retrieval, ensuring full transparency and reproducibility. The dataset covers a period from January 2010 through December 2024, with each observation representing the closing price of a given stock at the end of each month. By restricting attention to these market

leaders, the study can test predictive models in a segment characterized by deep liquidity and high informational efficiency.

Macroeconomic Variables: Federal Reserve Economic Data (FRED)

To complement firm-specific information, key macroeconomic variables were incorporated from the Federal Reserve Economic Data (FRED) repository via the fredapi library. The three macro indicators included are:

- **3-month Treasury Bill Rate (DTB3):** Used as a proxy for the risk-free rate and to capture changes in monetary policy stance.
- **Term Spread (T10Y2Y):** Measured as the difference between the 10-year and 2-year U.S. Treasury yields, indicating the slope of the yield curve and expectations for economic growth.
- **CBOE Volatility Index (VIXCLS):** Serves as a real-time gauge of market volatility and investor risk aversion.

Macroeconomic data were collected at a monthly frequency and merged with the stock panel by date, ensuring temporal alignment for each cross-sectional observation.

Data Processing and Feature Construction

Calculating Returns

Monthly simple returns were calculated for each stock as the percentage change in adjusted closing prices from one month to the next. These returns serve as the core dependent variable for the asset pricing models. Handling missing or incomplete data points was crucial; any months with missing returns due to non-trading or incomplete data were excluded to maintain a balanced panel.

Engineering Predictors

To enable meaningful predictions, a range of predictor variables was engineered, drawing on common practice in empirical asset pricing and the methodologies outlined by (Gu, Kelly, & Xiu, 2020). The following features were constructed for each stock and month:

- **12-Month Momentum:** The cumulative return over the preceding 12 months, excluding the most recent month, to capture medium-term price trends. Momentum is widely regarded as a key return predictor in finance.

- **3-Month Volatility:** The standard deviation of monthly returns over the last three months, reflecting the stock's recent risk profile.
- **Short-Term Reversal:** Defined as the negative of the previous month's return, designed to capture the well-documented reversal effect where recent losers outperform recent winners.
- **Past Return:** The lagged monthly return, included to directly test for short-term autocorrelation.

The **target variable** for modeling is the *next* month's realized return, making the setup a classic one-step-ahead cross-sectional forecasting problem.

Incorporating Macroeconomic Predictors and Interactions

After merging macroeconomic series to each (stock, month) observation, additional interaction terms were constructed between firm-level predictors (like momentum and volatility) and macroeconomic variables. This allows models to capture context-dependent effects—such as momentum signals being stronger during low-volatility or low-rate regimes. All predictors were standardized to ensure comparability and to aid model convergence in machine learning estimation.

Sector Dummies

Although the focus is on a handful of technology and consumer-facing giants, sector information was added using metadata from Yahoo Finance. Each sector was encoded as a dummy variable, providing flexibility for the models to account for residual differences in industry exposures.

Data Cleaning and Quality Assurance

Data cleaning steps were applied at several points to ensure robustness. Any observation with missing values in any predictor or target variable (often due to rolling-window calculations at the start of the sample) was dropped. Outlier checks were performed, particularly for calculated returns and macro variables, to ensure that extreme or spurious data points did not unduly influence model estimation. The final dataset is a panel where each row represents a (stock, month) pair, and each column is a standardized predictor or macroeconomic variable.

Methodology

Kolmogorov-Arnold Networks (KAN)

KAN is a recently developed neural network architecture grounded in the Kolmogorov-Arnold representation theorem, which states that any multivariate continuous function can be represented as a finite sum of univariate continuous functions. KANs are designed to efficiently approximate

complex, high-dimensional nonlinear relationships using adaptive basis functions.

A KAN model replaces the standard linear transformations between layers in a neural network with adaptive nonlinear transforms, learning a representation as:

$$f(x) \approx \sum_{q=1}^{Q} \phi_q\left(\sum_{p=1}^{P} \psi_{q,p}(x_p)\right)$$

where:

- $f(x) = f(x_1,\ldots\ldots,x_p)$: Input feature vector
- Q: Number of basis units (neurons)
- P: Number of input features
- $\psi_{q,p}(.)$: Learnable univariate nonlinear transformation of feature p at unit q (e.g., spline, ReLU)
- $\phi_q(.)$: Learnable nonlinear mixing function for basis unit q

In practical implementation, the pykan package offers users a high degree of flexibility, enabling them to configure the architecture by specifying the width (number of units per layer), the resolution of basis expansions through grid points, and the order of nonlinearity employed in the model. Model training

proceeds via standard supervised learning protocols, typically optimizing the mean squared error (MSE) between predicted and actual returns, thus aligning with established practices in machine learning regression.

Deep Neural Networks (DNN)

DNNs have become a standard tool in machine learning for tabular data with complex, nonlinear relationships. Their ability to capture interactions and nonlinearities makes them a natural candidate for empirical asset pricing, where return drivers are likely to interact in nontrivial ways.

A DNN is composed of multiple layers of neurons, each applying a linear transformation followed by a nonlinear activation function. For a simple feed-forward DNN with one hidden layer:

$$h = \sigma(W_1 x + b_1)$$

$$\hat{y} = W_2 h + b_2$$

where:

- x : input vector (predictors)

- W_1, W_2: weights
- b_1, b_2: biases
- $\sigma(.)$: activation function (e.g., ReLU, tanh)

The model is trained to minimize the loss:

$$L(\theta) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

where:

- $L(\theta)$: The loss function
- N: Total number of data samples (training observations)
- y_i: True value (actual target) for the *i*-th sample
- $\hat{y}_i$: Predicted value for the *i*-th sample, given by the model
- i: Index running over all samples from 1 to N

Recurrent Neural Networks (RNN)

RNNs, especially Long Short-Term Memory (LSTM) variant, are designed to model temporal dependencies by retaining information from previous time steps. While less common for pure cross-sectional prediction, I included RNNs

to test whether temporal dynamics in features across months for each stock could improve forecasting.

The core of an RNN is its recursive update:

$$h_t = \sigma(W_h h_{t-1} + W_x x_t + b)$$

$$\hat{y}_t = W_y h_t + b_y$$

where:

- h_t: Hidden state at time t
- σ: Activation function (e.g., tanh or ReLU)
- W_h: Weight matrix for the previous hidden state
- W_x: Weight matrix for the current input
- x_t: Input feature vector at time t
- b: Bias vector for the hidden layer
- $\hat{y}_t$: Model output (prediction) at time t
- W_y: Output layer weights
- b_y: Output layer bias

For the cross-sectional asset pricing setting, the recurrent neural network (RNN) is structured to process each stock's sequence of monthly feature

vectors as its input. By ingesting the historical progression of characteristics for a given stock, the RNN leverages temporal patterns and dependencies present in the data. At each time step, the network produces a prediction for the following month's return, conditioning on both current and past information. This approach allows the model to incorporate information from the entire feature history of each stock, potentially uncovering dynamic relationships that may improve the accuracy of out-of-sample forecasts.

Gradient-Boosted Regression Trees (GBRT)

GBRT, such as XGBoost or LightGBM, is among the most successful methods for structured/tabular data, including in asset pricing. Trees naturally capture nonlinearity and interaction effects, and boosting combines many weak learners to form a strong overall predictor.

A GBRT model builds a sequence of decision trees, where each new tree is trained to predict the residual error of the combined previous trees. The final prediction is:

$$\hat{y}_i = \sum_{m=1}^{M} f_m(x_i)$$

where f_m is the *m*-th tree.

Boosting minimizes the loss:

$$L(\theta) = \sum_{i=1}^{N} l(y_i, \hat{y}_i)$$

with l often being squared error.

Ordinary Least Squares (Linear Regression)

The baseline for asset pricing is the linear cross-sectional regression, which regresses next month's return on all available predictors. This model provides a transparent and interpretable benchmark.

$$y_i = \alpha + \sum_{j=1}^{p} \beta_j x_{ij} + \varepsilon_i$$

where:

- y_i: Next month's return for stock i
- x_{ij}: Value of predictor j for stock i
- β_j: Estimated coefficient for predictor j
- α: Regression intercept
- ε_i: Error term for stock i

- p: Number of predictors

OLS solves for β to minimize the sum of squared errors.

Out-of-Sample Evaluation and Portfolio Construction

Out-of-Sample R^2

To assess the **statistical predictive power** of each model, we calculate the out-of-sample R^2:

$$R^2_{\text{oos}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i \left(y_i - \overline{y}_{\text{train}}\right)^2}$$

where:

- y_i: Actual return in the test set
- $\hat{y}_i$: Predicted return for the i-th observation
- $\overline{y}_{\text{train}}$: Mean return in the training set

A positive R^2 indicates superior prediction to a naive mean-forecast; negative values suggest overfitting or lack of signal.

Portfolio Sorting and Long-Short Strategies

To move from statistical metrics to economic relevance, we implement a portfolio sorting strategy each month, as done in (Gu, Kelly, & Xiu, 2020):

1. For each month, rank stocks by their predicted return from each model.
2. Assign stocks to deciles (top 10%, bottom 10%).
3. Construct a long-short portfolio: Long the top decile, short the bottom decile.

Sharpe Ratio

The Sharpe ratio **measures the average excess return per unit of risk:**

$$\text{Sharpe Ratio} = \frac{E[r_p - r_f]}{\sigma(r_p)} \times \sqrt{12}$$

A higher Sharpe ratio reflects a better risk-adjusted return and is the gold standard for comparing strategies in finance.

Maximum Drawdown

The **maximum drawdown (DD)** quantifies the largest observed loss from a historical peak to a trough before a new peak:

$$\text{Drawdown}_t = \frac{P_t - \max_{s \le t} P_s}{\max_{s \le t} P_s}$$

$$\text{Max Drawdown} = \min_t(\text{Drawdown}_t)$$

Where, P_t: Cumulative portfolio value at time t

Maximum drawdown measures tail risk and investor pain—strategies with low Sharpe but large drawdowns are unattractive, even if mean returns are high.

Empirical Findings

This section presents the empirical results from applying a suite of machine learning models—including Kolmogorov-Arnold Networks (KAN), deep neural networks (DNN), recurrent neural networks (RNN), gradient-boosted regression trees (GBRT), and linear regression—to the monthly cross-section of the Magnificent 7 U.S. stocks using features derived from Yahoo Finance and macroeconomic data from FRED. We evaluate model performance along three dimensions: statistical fit (out-of-sample R^2), economic value (monthly long-short Sharpe ratio), and portfolio risk (maximum drawdown). These results are then contrasted with the benchmark findings of Gu, Kelly, and Xiu (2019), who examine a much broader panel of over 30,000 stocks and nearly a thousand predictors over 60 years, and who report substantially higher out-of-sample predictability and economic gains for nonlinear models, especially in more heterogeneous stock universes.

The table below summarizes the out-of-sample performance metrics for each model.

Model	Out-of-sample R^2	Sharpe Ratio	Max Drawdown

Kolmogorov-Arnold Network (KAN)	- 4.11	0.02	-48.91%
Deep Neural Network (DNN)	-7.95	-0.90	-89.34%
Recurrent Neural Network (RNN)	-0.68	-0.92	-82.71%
Gradient-Boosted Regression Trees (GBRT)	-1.50	0.74	-43.17%
Ordinary Least Squares Linear Regression (OLS)	-0.32	-0.63	-85.39%

Compared to the machine learning benchmarks set in (Gu, Kelly, & Xiu, 2020), where neural networks and tree-based methods achieved *positive* out-of-sample R^2 values (often 0.3%–0.4% per month for stock-level prediction and up to 1–1.8% for aggregated portfolios) and annualized long-short Sharpe ratios as high as 1.35, our models display notably weaker predictability and economic

value. For the Magnificent 7, all models exhibit negative out-of-sample R^2, indicating that none could improve upon the naive mean forecast, and only GBRT yields a Sharpe ratio above zero, suggesting that nonlinearity still offers a modest edge in extremely concentrated and efficient stock universes. Maximum drawdowns further highlight the difficulty of constructing robust portfolios from such a narrow sample, with most strategies experiencing significant declines, in contrast to the more diversified and resilient results documented in broader cross-sections by (Gu, Kelly, & Xiu, 2020).

These findings underscore several important points. First, the high cross-sectional efficiency and low return dispersion among the largest U.S. stocks make statistical prediction exceptionally difficult, and traditional as well as sophisticated machine learning methods struggle to identify persistent alpha. Second, economic metrics such as Sharpe ratios and drawdowns remain crucial for evaluating model effectiveness in a portfolio context—mere statistical fit is insufficient in practical investment scenarios. Lastly, our results align with the central insights of the (Gu, Kelly, & Xiu, 2020) paper: machine learning's economic value is most evident in broad, noisy, and heterogeneous universes, where nonlinearities and interactions are more pronounced and exploitable. In contrast, narrow and liquid segments like the

Magnificent 7 present a much more challenging environment for return prediction.

Interpretation of Cumulative Return Profiles

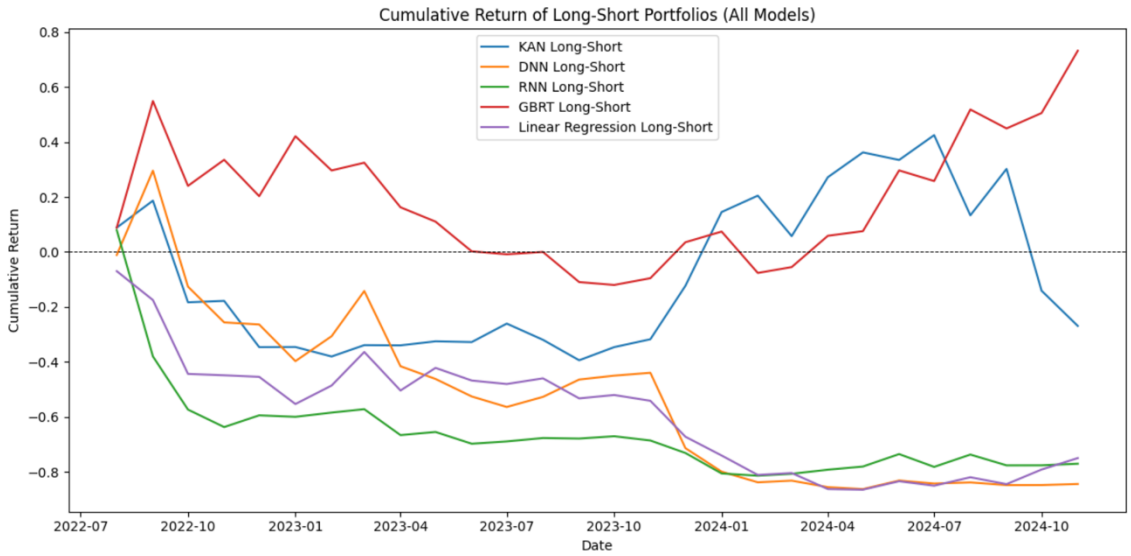

Figure 1.0: Cumulative Return of Long-Short Portfolios (All Models)

Figure 1.0 presents the cumulative returns of monthly long-short portfolios constructed using predictions from each machine learning model. The graph makes clear that, over the test period, most models struggled to generate persistent, positive excess returns. While the GBRT-based long-short portfolio achieved a consistent upward trajectory and finished with the highest cumulative return, all other models exhibited flat or declining performance, with deep and recurrent neural networks (DNN, RNN) and linear regression experiencing sustained losses. The KAN model showed episodes of positive performance but failed to deliver stable gains, highlighting the challenges of capturing cross-sectional alpha in a highly efficient, narrow sample of large-cap stocks.

These results reinforce the earlier statistical findings: even with sophisticated nonlinear models, extracting meaningful predictive signals from the Magnificent 7 universe is extremely challenging. The generally negative or flat cumulative returns across models underline the low cross-sectional dispersion and efficiency present among these stocks, in stark contrast to results reported in broader equity universes (such as in (Gu, Kelly, & Xiu, 2020)). Importantly, while Sharpe ratios and drawdown metrics provide summary statistics, the cumulative return paths here visually demonstrate the practical difficulty of

building successful long-short strategies in this environment. Only GBRT managed to avoid significant drawdowns and deliver net positive returns, suggesting that some tree-based nonlinearities may still be exploitable—but the overall takeaway is that model choice alone is insufficient to overcome the limitations imposed by data scope and market structure.

Conclusion

This study set out to evaluate the predictive and economic performance of a range of machine learning models—including Kolmogorov-Arnold Networks (KAN), deep and recurrent neural networks (DNN, RNN), gradient-boosted regression trees (GBRT), and linear regression—within the challenging

environment of the Magnificent 7 U.S. stocks. Despite leveraging advanced modeling techniques and following the out-of-sample R^2 evaluation methodology of (Gu, Kelly, & Xiu, 2020), all models yielded negative out-of-sample R^2, indicating that none could beat the naive mean forecast. This stands in contrast to the original (Gu, Kelly, & Xiu, 2020) paper, where out-of-sample R^2 values up to 1% were achieved on a far richer and more diverse cross-section of U.S. equities using a broader feature set and larger sample.

Several factors contribute to these results. The dataset's limited cross-sectional breadth—just seven highly efficient, liquid stocks—along with a relatively small sample size and reliance on simple, mostly price-based features, imposed strict limits on the potential for uncovering return predictability. Additionally, the noisy and low-dispersion nature of returns among the Magnificent 7 made the statistical challenge even greater. While GBRT achieved a modestly positive Sharpe ratio in portfolio backtesting, and economic evaluation through long-short decile spreads was performed in line with best practices, the overall ability of machine learning models to generate economically significant signals in such a restricted universe remained weak.

In summary, this analysis reinforces the core insight of (Gu, Kelly, & Xiu, 2020): the power of modern machine learning methods in asset pricing is most

evident in broad, heterogeneous samples with rich predictor sets. For extremely concentrated, large-cap universes with limited features, statistical and economic predictability remains elusive. Future research should focus on expanding data richness—both in the cross-section of assets and the diversity of features—to unlock the full potential of nonlinear modeling for empirical asset pricing.

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